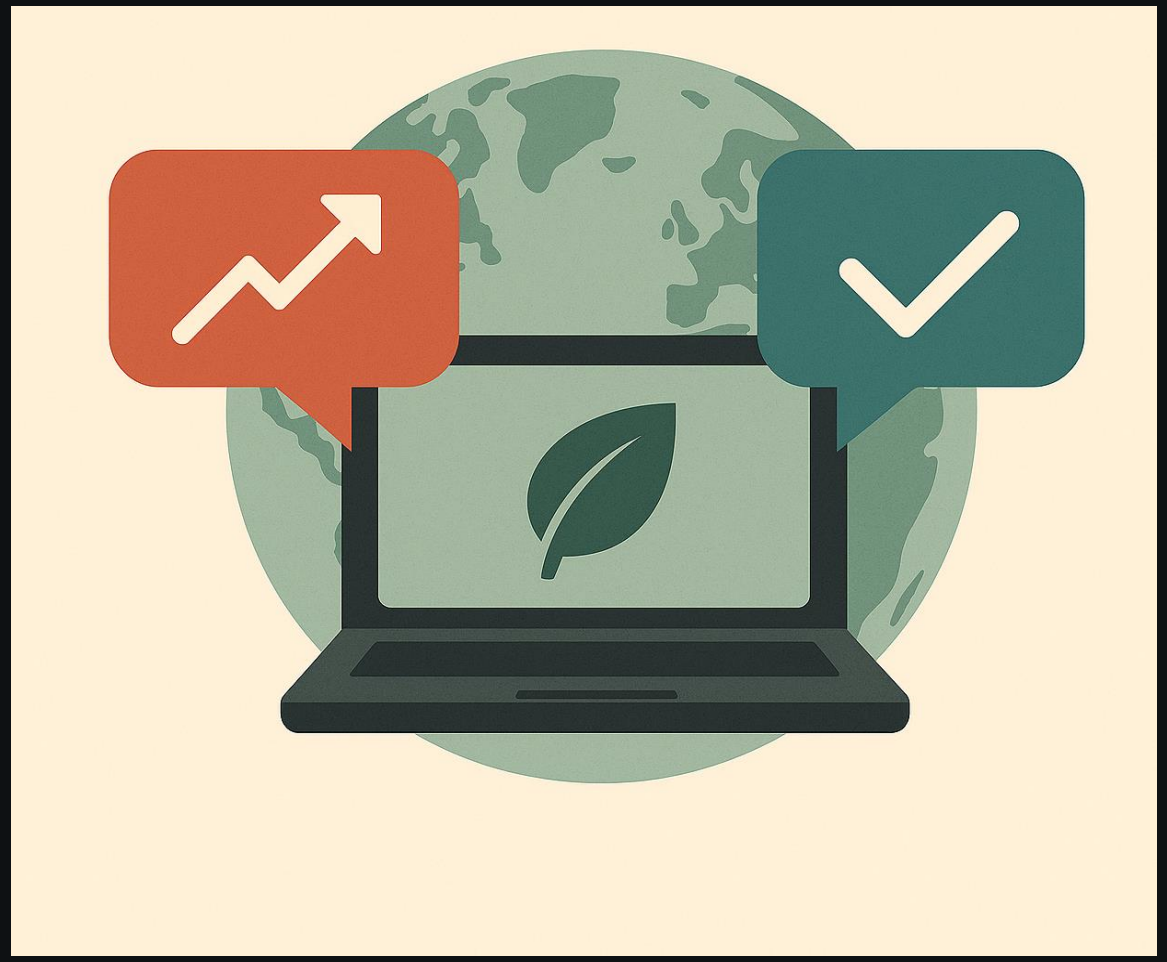
r/macss30112 • 5d ago
ClimateBias

Analyzing the relationship between political tone, accuracy, and engagement in online discourse

By Jacob Rampino, Emely Antunes Otsuka, Kayla Hammonds.

101 · 23

Project Overview



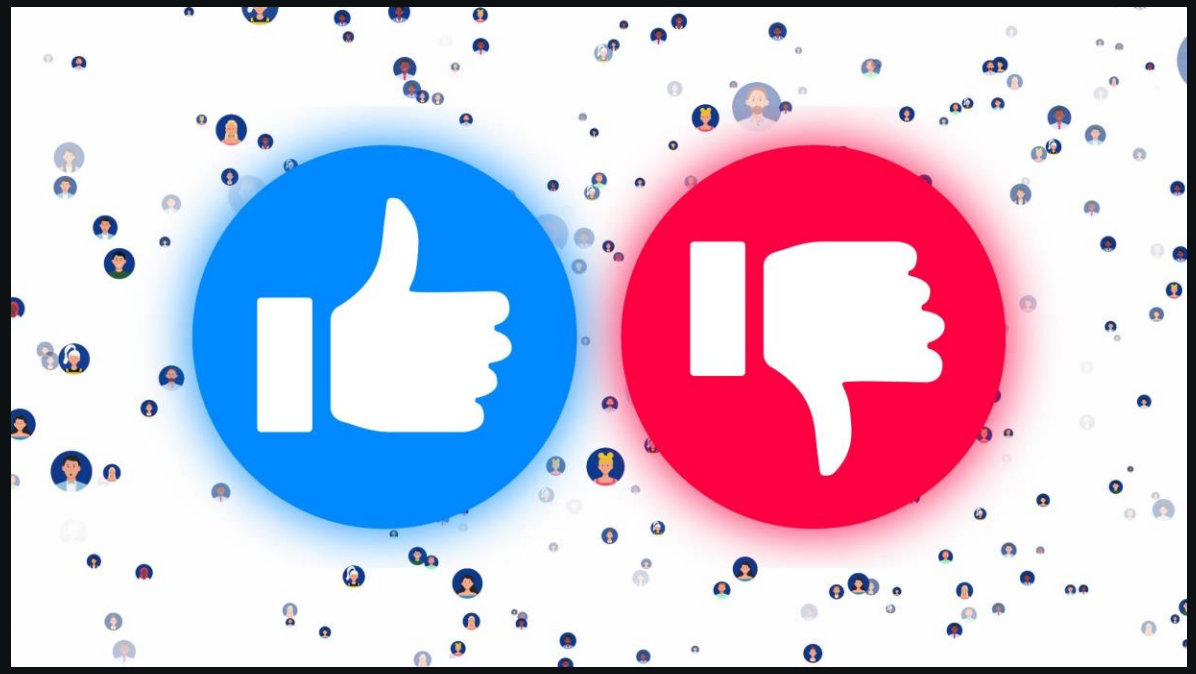
Project Objective

The project studies how partisan language and factual accuracy influence engagement in climate discussions online.

Analysis Focus

The study maps interactions among political leaning, factual credibility, and community responses like karma scores.

Research Question



Central Research Question

Examines how partisanship strength and accuracy of online statements influence social approval across communities.

Significance of Online Discourse

Online discourse shapes public climate policy views, with social media as key political engagement venues.

Political Distortion in Climate Communication

Climate science is often reframed through partisan lenses, amplifying **division** and undermining **trust**.

Expected Research Insights

Left-leaning and accurate statements likely to get higher engagement; right-leaning patterns may differ.

Data Sources and Collection Methods

SOURCE	SAMPLE SIZE	TIME PERIOD	COLLECTION METHOD	KEY VARIABLES
Reddit Climate Dataset	- Large multi-topic dataset - 83,938 samples with 14 features	Varies by subreddit but ranges from 2008 to 2023	Downloaded Dataset from IEEE Dataport	Engagement rate, tokenized text
Congressional Speeches	- 1.3K floor speeches	2017 to 2022	API extraction from Congress.gov and GovInfo	Weighted lexical scores by party
Wikipedia Articles	- 50 climate-related pages	N/A	Web scraping	Paragraph chunks

Data Cleaning

- **Reddit:** missing fields, calculate engagement, zero community members, combine datasets
- **Wikipedia:** filtering out loosely related pages, different languages, in-text citations and references
- **Congress:** parsing records for speeches, dropping procedural language, and identifying speaker party.

One Takeaway/ Challenge

- **Parsing Congressional Record HTMLs!** The text of these reports are all placed in one giant tag, meaning speech parsing is entirely regex based across 6 years of data.

```
https://www.congress.gov/115/crec/2017/02/09/modified/CREC-2017-02-09-pt1-PgH1094-2.htm,"EXPENDITURE REPORTS CONCERNING OFFICIAL
<head>
<title>Congressional Record, Volume 163 Issue 23 (Thursday, February 9, 2017)</title>
</head>
<body><pre>
[Congressional Record Volume 163, Number 23 (Thursday, February 9, 2017)]
[House]
[Pages H1094-H1098]
From the Congressional Record Online through the Government Publishing Office [<a href=""https://www.gpo.gov"">www.gpo.gov</a>]




        EXPENDITURE REPORTS CONCERNING OFFICIAL FOREIGN TRAVEL

  Reports concerning the foreign currencies and U.S. dollars utilized
for Official Foreign Travel during the fourth quarter of 2016, pursuant
to Public Law 95-384, are as follows:

                REPORT OF EXPENDITURES FOR OFFICIAL FOREIGN TRAVEL, COMMITTEE ON AGRICULTURE, HOUSE OF REPRESENTATIVES, E
-------------------------------------------------------------------------------------------------------------------------
                                                  Date                                    Per diem \1\          Transpor
                                        ---------------------                     ----------------------------------
                                                                                                  U.S. dollar
      Name of Member or employee                                    Country        Foreign     equivalent    Foreign
                                        Arrival   Departure                        currency     or U.S.      currency
                                                                                               currency
-------------------------------------------------------------------------------------------------------------------\2\-------
Hon. Conaway............................  10/16      10/18   Egypt..................  ...........      586.98  ...........
                                           10/18      10/20   Kuwait.................  ...........      787.09  ...........
                                           10/20      10/23   Israel.................  ...........    1,934.00  ...........
                                           10/23   ..........  USA....................  ...........  ...........  ...........
```

Data Validation and Quality Checks

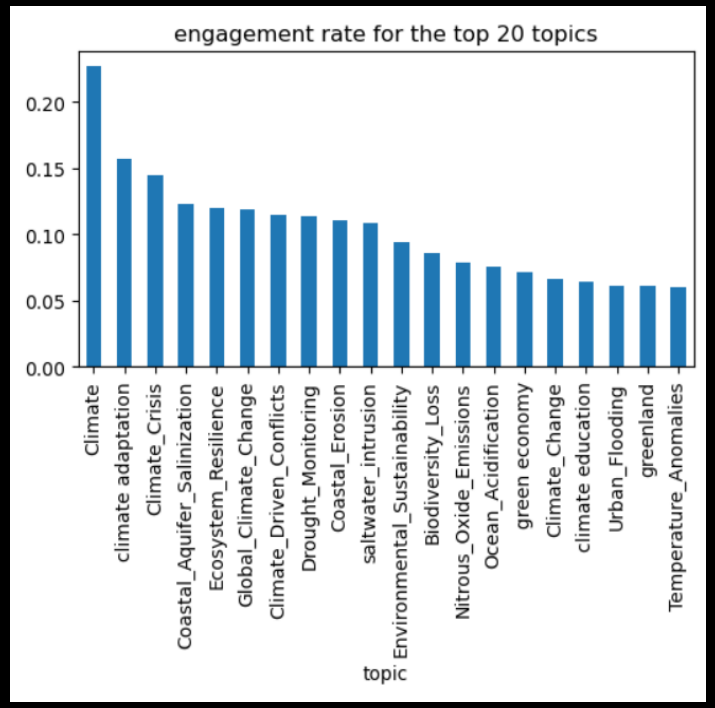
- **Reddit:** The text collected from the dataset downloaded was scraped by the authors by collecting using 52 climate-related keywords to identify relevant posts across communities. Therefore, we trust the dataset to return only relevant posts. However, as seen by the title of the post with the least context similarity to the Wikipedia chunks (slide 8), the text belonged to the topic of “wildfires” even if the text had no mentions of it. Thus, we have two hypothesis that can explain this discrepancy: the first is a simple error in the API used by the authors; or that the deleted photo in the photo was relevant to the topic of wildfires. Either way, since our investigation matched text with chunk, a small quantity of mis-collected posts would still return a low similarity and not affect our scores as much.

- **Wikipedia:** Wikipedia was used as a measure of not only neutrality but also factual language. Since climate change is a political topic that has a major scientific component to it, we wanted the source of our accuracy to reflect objective and consensus-based language backed by citations. With this in mind, we chose Wikipedia for two reasons: the collaborative nature of the platform, their policies on citations and maintaining a neutral point of view, and also being a relatively easy website to scrape.

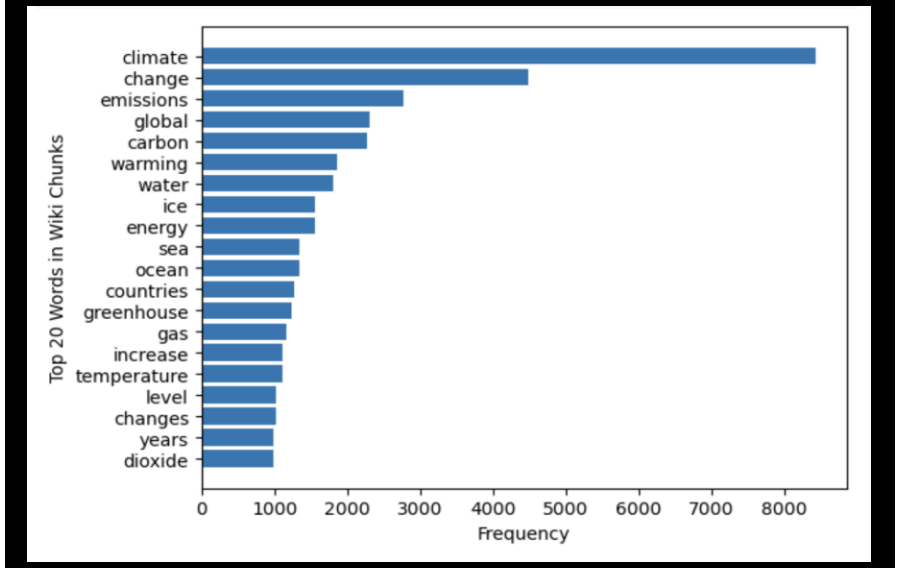
- **Congress:**
  - The Congressional Dataset required two major checks.
    - Foremost, I had to verify the text collected was not procedural and was in fact a floor speech. This required manually going through collected speeches and verifying they were not procedural, and if they were, updating the regular expressions to account for those situations. I trust the included speeches are speeches, but it is very plausible they are an undercount as many extremely short 'speeches' (such as passing time to another member) would likely be excluded.
    - Second, I had to verify the lexical scores were coherent, and I did so by printing the most weighted words for each party. Because the model does not encode semantics, these words were coherent, but individual phrase scoring is not always perfect because it cannot understand the context in which words are used. This is an inherent limitation of this model but is expected and still largely robust for parsing similarity to a certain party's speech.

Individual Distributions

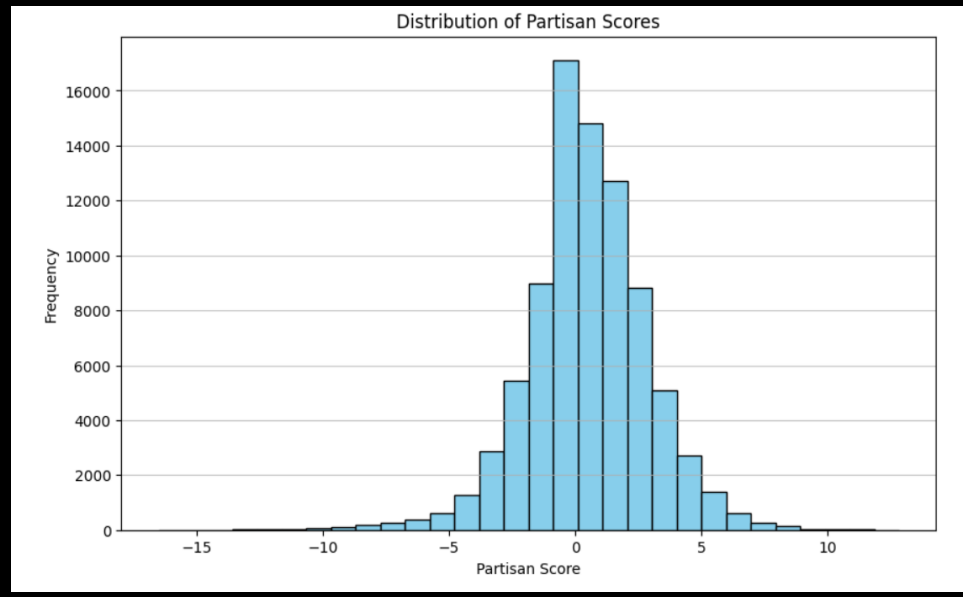
Reddit Posts



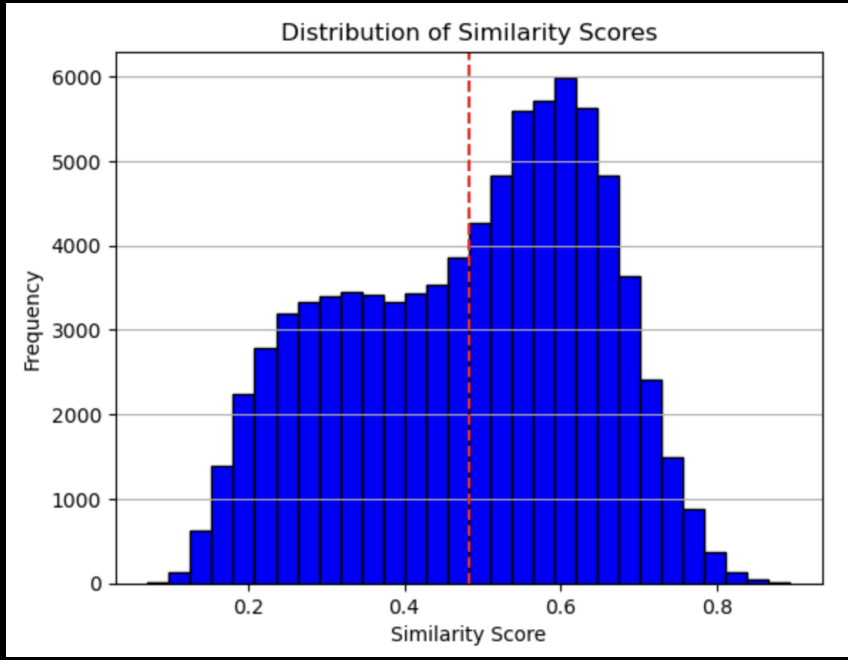
Wikipedia Pages

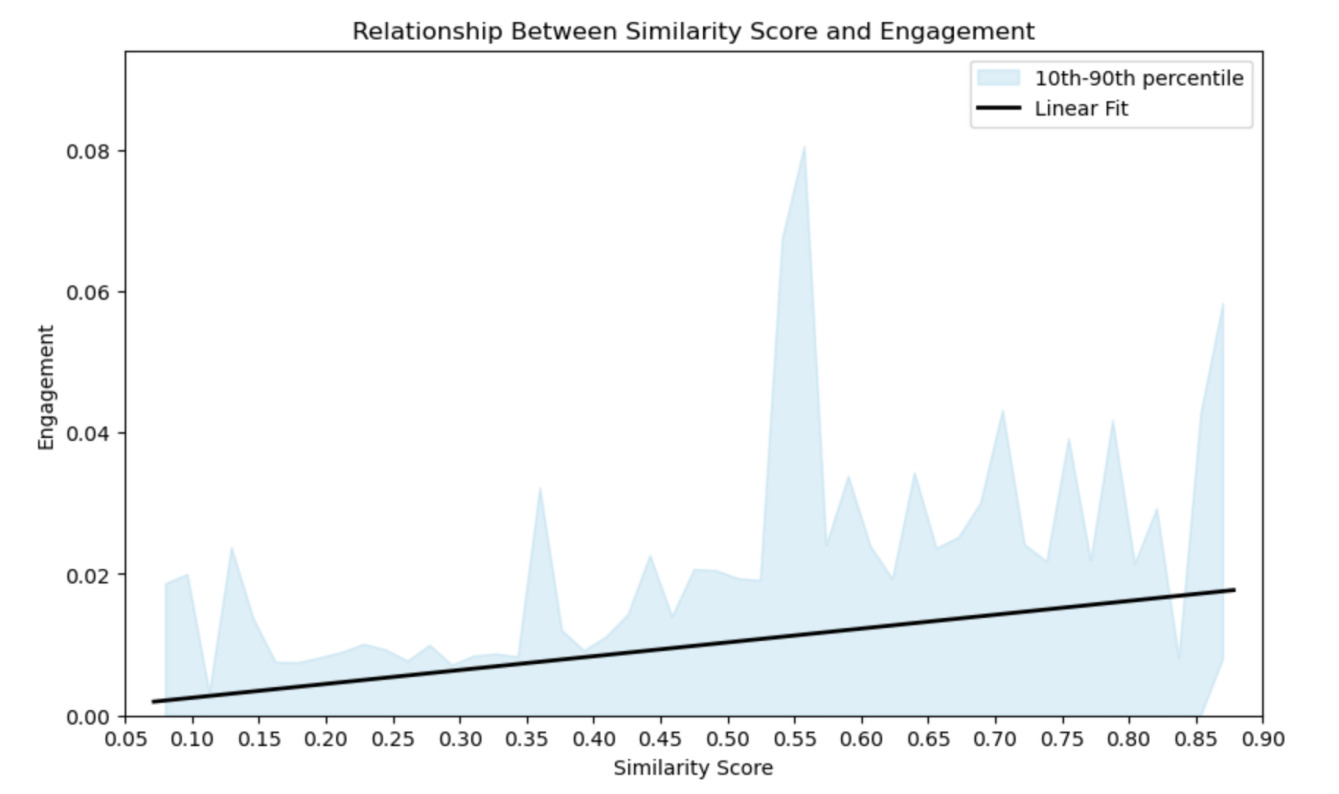


Reddit Partisanship



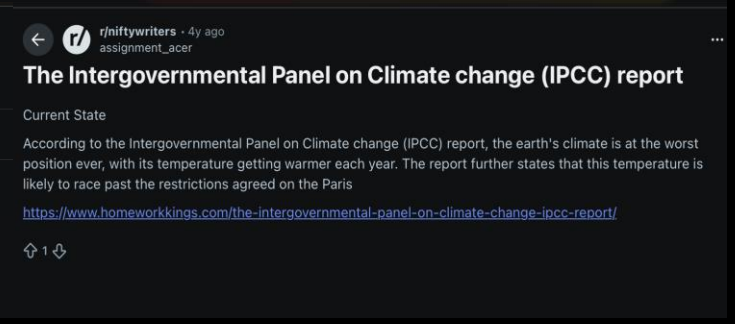
Engagement Rate x Similarity Score





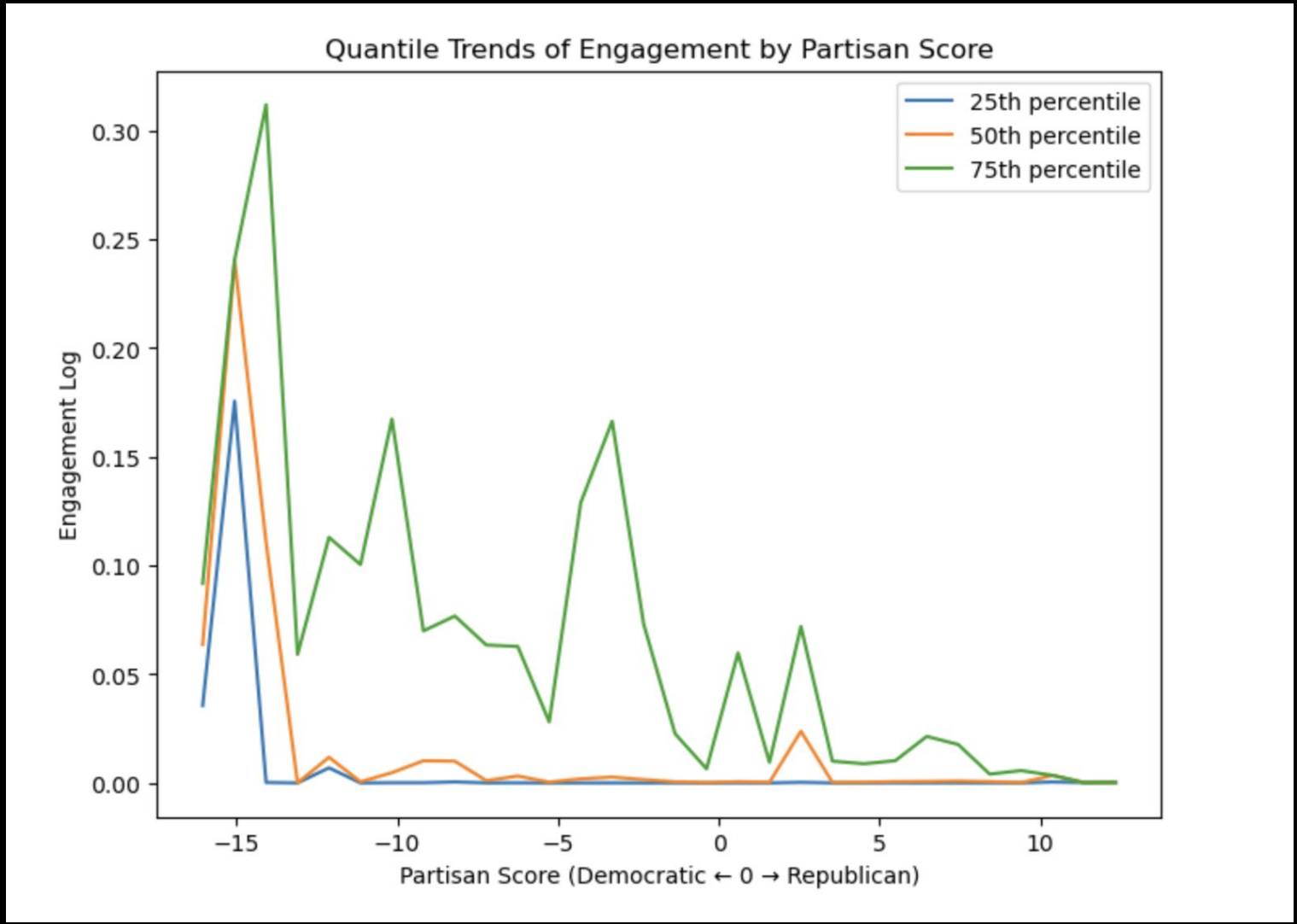
Lowest = 0.07

Highest = 0.89



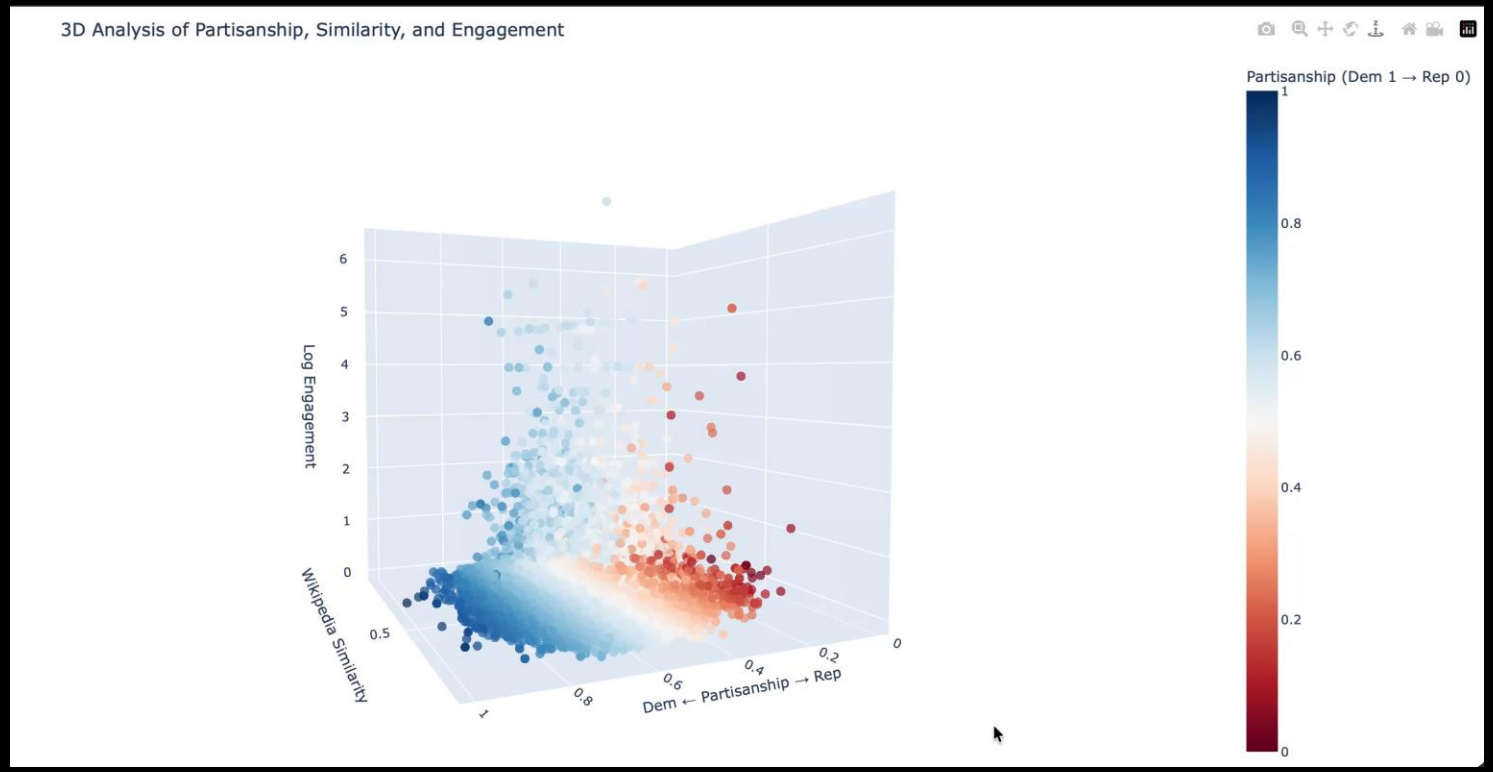
3/10/2026

Engagement Rate x Partisanship Score



3/10/2026

Engagement Rate x Similarity Score x Partisanship Score



Click to explore plot

3/10/2026

Conclusion

Mixed Effects Interpretation

- **Key insight**:
  - **The effect of similarity on engagement changes depending on ideology**
    - **Dem-leaning**
      accuracy boosts engagement
    - **Rep-leaning**
      moderates accuracy's effect on engagement
- Partisanship alone is **not** significant (coef. = 0.052, p = 0.279)

```
                     Mixed Linear Model Regression Results
==============================================================================
Model:                    MixedLM      Dependent Variable:      engagement_log_x
No. Observations:         140946       Method:                  REML
No. Groups:               52           Scale:                   0.1435
Min. group size:          158          Log-Likelihood:          -63463.0974
Max. group size:          18187        Converged:               Yes
Mean group size:          2710.5
------------------------------------------------------------------------------
                                      Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------------------------
Intercept                            -0.209    0.104 -2.002 0.045 -0.414 -0.004
similarity_score                      0.907    0.253  3.580 0.000  0.411  1.404
partisan_normalized                   0.052    0.048  1.083 0.279 -0.042  0.145
similarity_score:partisan_normalized -0.479    0.090 -5.310 0.000 -0.656 -0.302
Group Var                             0.526    0.277
Group x similarity_score Cov         -1.291    0.681
similarity_score Var                  3.191    1.679
==============================================================================
```

We find that accuracy (similarity to Wikipedia) of a post does improve engagement if language also leans towards Democratic lexical features, but accuracy has no significant effect if a post leans towards Republican lexical features.